

National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	PF (Lab)	Course Code:	CL 1002
	Program:	BS (CS)	Semester:	Fall 2024
	Duration:	2 Hours	Total Marks:	30 (3*10)
	Paper Date:	14-Oct-2024	Weight	30%
	Section:	C	Page(s):	4
	Exam:	Lab Midterm Exam	Reg. No	

Important Instructions (Please read them before attempting the exam):

- Submit **ONLY .cpp File** in this format (Make the File **named** with your **Roll Number** e.g., 24L_1234.cpp).
- **Plagiarism** will result in **F grade** in lab.
- No cell phones are allowed. Sharing of **USBs** or any other items is **not allowed**.

Question # 1:

A bank offers various loan types to customers based on their credit score and loan amount. The loan system is tiered as follows:

- **Loan types:** Personal, Auto, Mortgage
- **Credit score ranges:**
 - Excellent (750+): 4% interest.
 - Good (600-749): 6% interest.
 - Fair (450-599): 8% interest.
 - Poor (<450): Loan denied.
- **Loan amount:**
 - Personal loans: Max \$50,000.
 - Auto loans: Max \$100,000.
 - Mortgage: Max \$500,000.

Write a program that:

- Takes as input the loan type, credit score, and requested loan amount.
- Based on the user's credit score and loan type, determine the loan eligibility and the interest rate using **nested if-else** structures.
- If the requested loan exceeds the allowed amount for the selected type, deny the loan and inform the user.
- Handle different loan types and validate input using a **do-while loop** to ensure the loan type, credit score, and loan amount are valid.

- If the loan is approved, calculate the monthly payment using the formula for simple interest over a variable term (1-30 years), which the user should input.
- Use if-else-if structures to apply dynamic interest rate changes based on government regulations. For example, after \$200,000 in loans, add an extra 1% to the rate.

Sample Input & Output (Approved Loan):

Enter loan type (1 for Personal, 2 for Auto, 3 for Mortgage): 3

Enter your credit score: 780

Enter the requested loan amount: 250000

Enter the loan term in years (1 to 30): 20

Loan Type: Mortgage

Credit Score: Excellent (780)

Requested Loan Amount: \$250000

Loan Approved!

Interest Rate: 4%

Dynamic Interest Rate after \$200,000: 5%

Loan Term: 20 years

Monthly Payment: \$1458.33

Question # 2:

A factory operates a precision-controlled production line, which manufactures high-tech components. Each component has strict quality control rules, and the production process involves multiple steps.

Each batch of components produced must meet the following criteria:

- Every component's weight must be within a precision range (between 0.999 kg and 1.001kg).
- The production system allows the user to enter the number of components and their weights.
- The system should reject and flag any components that fall outside the precision range.

Write a program that:

- Uses a for loop to process each component in a batch.
- For each component, use if-else to check whether the weight is within the allowed range.
- Count and report the number of components outside the acceptable precision range.
- Use a do-while loop to validate the input of component weights (ensuring they are positive).
- After processing a batch, allow the user to continue with another batch or quit.

- Calculate the average weight deviation of flagged components and report it.
- Include penalties for more than 5 rejected components in a batch (use a while loop to add cumulative penalties and adjust the total number of acceptable components).
- if a session cannot be accommodated in the day due to time or capacity constraints, move it to the next available day or room and adjust the schedule accordingly.
- At the end of the scheduling process, print a summary for each day, showing the number of sessions, total duration, and unused time slots for each room.

Sample Input & Output:

Enter the number of components in the batch: 7

Component 1 weight: 1.005

Component 2 weight: 0.995

Component 3 weight: 1.006

Component 4 weight: 0.998

Component 5 weight: 0.999

Component 6 weight: 1.007

Component 7 weight: 0.997

Rejected components: 5 (Component 1, Component 2, Component 3, Component 6, Component 7)

Average weight deviation of rejected components: 0.004 kg

Penalty applied: Total acceptable components reduced by 1.

Do you want to process another batch? (Y/N): N

Question # 3:

A company organizes conferences that span multiple days, with each day having multiple sessions. Each session can have a varying number of attendees and duration. The system needs

to optimize the schedule by ensuring that:

- The total duration of all sessions in a day does not exceed 10 hours.
- The number of attendees per session should not exceed the room capacity.

Write a program that:

- Uses **nested loops** to handle multiple days and multiple sessions for each day.
- For each session, input the duration and number of attendees.
- Use a **while loop inside a for loop** to track the total hours for each day, ensuring it does not exceed 10 hours.
- Use a **do-while loop** to validate that the number of attendees does not exceed the room capacity (input is invalid until this condition is met).
- if a day exceeds 10 hours or a session exceeds the capacity, provide a message and force re-entry.

- The system should handle multiple rooms per day, where different sessions can be assigned to different rooms. Use **nested while loops** to iterate over room assignments for each session.

Sample Input & Output (Day with 2 Rooms, 3 Sessions):

Enter the number of days for the conference: 1
Enter the number of rooms available for each day: 2

--- Day 1 ---

Enter details for Room 1:
How many sessions will be scheduled in Room 1? 2

Session 1:
Enter duration of session (hours): 3
Enter number of attendees: 25
Capacity of Room 1: 30
Session added successfully.

Session 2:
Enter duration of session (hours): 8
Total duration exceeds the 10-hour limit for the day. Please re-enter the session duration.
Enter duration of session (hours): 7
Enter number of attendees: 40
Attendees exceed the room capacity. Please re-enter the number of attendees.
Enter number of attendees: 28
Session added successfully.

Enter details for Room 2:
How many sessions will be scheduled in Room 2? 1

Session 1:
Enter duration of session (hours): 4
Enter number of attendees: 20
Capacity of Room 2: 25
Session added successfully.
Summary for Day 1:
- Room 1: 2 sessions scheduled, total duration = 10 hours, no unused time.
- Room 2: 1 session scheduled, total duration = 4 hours, unused time = 6 hours.